

FRESHNESS CANON

2026-03-18

Dexter Hadley, MD/PhD

Hadley Lab CANONIC

Abstract

Unchanged sources produce identical outputs. Skip the work. Trust the cache.

hadleylab.org Governed Research. Every claim cited.

Contents

service = FRESHNESS
primitive = INTEL
function = CACHE

Constraints

- MUST: Source dependency rules are declared in FRESHNESS
- MUST: TOOLCHAIN hash change invalidates entire cache
- MUST: CI always runs --no-cache (correctness is the goal)
- MUST: Cache is a runtime artifact at ~/.canonic/build
- MUST: MONITORING deep health reports compile freshness
- MUST: Freshness decisions emit structured output (JSON)
- MUST NOT: Skip validation after incremental build 255
- MUST NOT: Skip GC pass orphan cleanup always executes
- MUST NOT: Cache stale outputs missing output files for

FRESHNESS | CANON | TOOLCHAIN